Software Quality Assurance Practices Towards Waterfall and Agile Information Systems Projects of BOI Registered Software Companies in Sri Lanka

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***Abstract*—Software quality assurance (SQA) practices play a significant role to deliver high quality software product by systematically monitoring and evaluating the entire software development life cycle. In Sri Lanka's emerging software industry should focus on the proper SQA practices. Unfortunately a large number of companies fail to achieve their project objectives due to poor SQA practices. To avoid such situations, it is important to identify key factors in SQA practices and take appropriate SQA practices. In this context, SQA practices are important to identify defects early, reducing post-release costs and ensuring software reliability, performance, and compliance with industry standards, ultimately enhancing user satisfaction, mitigating risks, and fostering sustainable growth and customer retention. Hence the main objective of this paper is to identify SQA practices towards Waterfall and Agile information systems (IS) projects of BOI registered software companies in Sri Lanka. Further, this paper helpful for identification of key factors in SQA practices for Waterfall and Agile IS projects of BOI registered software companies in Sri Lanka, while being able to identify challenges, identifies of SQA practices and provides valuable recommendations.**

Keywords—Agile, BOI, information systems projects, software quality assurance practices, Waterfall

I. INTRODUCTION

SQA practices represent a comprehensive approach that involves continuous validation of quality. This encompasses the verification processes incorporated within the software development lifecycle [10,11].

A. Problem

The study identified several key problems regarding SQA towards Waterfall and Agile IS projects of BOI registered software companies in Sri Lanka. There is a limited understanding of the SQA practices adopted for Waterfall and Agile IS projects, insufficient comprehension of the key factors guiding the identification of SQA practices, and a lack of insights into the specific challenges and constraints faced in this context. Additionally, there is a need for comprehensive recommendations to align SQA practices with project methodologies and global industry standards.

The study identified a significant research gap by focusing a key problem statement:

"There is a notable absence of comprehensive research that identifies SQA practices towards Waterfall and Agile IS projects of BOI registered software companies in Sri Lanka."

B. Research objectives

a) General objective

The general objective of this paper is to: "To identify SQA practices towards Waterfall and Agile IS projects of BOI registered software companies in Sri Lanka."

b) Specific objectives

The specific objectives of this paper are to: (1) to identify SQA practices towards Waterfall and Agile IS projects in global context, (2) to identify key factors of SQA practices towards Waterfall and Agile IS projects of BOI registered software companies in Sri Lanka, (3) to identify challenges of SQA practices towards Waterfall and Agile IS projects of BOI registered software companies in Sri Lanka.

II. LITERATURE REVIEW

The Waterfall methodology works best when project criteria are established and explicit, the technology environment is static and well-understood, considerable resources and skills are available, and the project time is short, resulting in methodical development and successful outcomes [14].

The Agile methodology promotes iterative development and flexibility, breaking down projects into manageable sub-modules for stakeholder assessment with each iteration. However, it necessitates rigorous limits to avoid making excessive recommendations and may confront difficulties with stakeholder participation and large-scale project organization [15].

The size and complexity of a software development project frequently affect the methodology used. Agile is ideal for small projects because of its flexibility, whereas Waterfall is ideal for large, complex projects. However, medium-sized projects may require hybrid techniques due to the constraints of both methodologies [1,15]. The importance of quality in software engineering is generally recognized. Agile methodologies include quality-focused processes into the development lifecycle, allowing for faster production of high-quality software that adapts to changing client requirements. Numerous scholarly studies have investigated software quality within Agile methodologies [2]. As software development businesses shift to Agile methodologies, it is critical to reevaluate and restructure human resource requirements and team structures. Unlike the Waterfall process, Agile necessitates a collaborative approach in which varied skill sets come together within project teams. Emphasizing human abilities, particularly successful cooperation and communication, becomes critical [14].

Waterfall and Agile methodologies face challenges such as insufficient training, communication barriers, and cultural disparities. Effective methodology adoption requires organizational change, early process formulation, proper structures, pilot projects, ongoing evaluation, and top management support. Strategic planning includes training, facilitators, frameworks, and coaching. Cultural compatibility, readiness, and alignment are all important considerations [8].

A. *Key factors of SQA practices towards Waterfall IS projects*

The following table presents the identified key factors of SQA practices for Waterfall IS projects.

TABLE 1. KEY FACTORS OF SQA PRACTICES TOWARDS WATERFALL IS PROJECTS

Categories	Factors	Research source
Project	Project type	[1], [2], [4], [6], [8], [9], [10], [11], [13], [18]
	Project size	
	Project complexity	
	Project timeline	
	Project cost	
	Project scope	
	Resource availability	
Stakeholder	Requirement	[1], [4] [6], [8], [9], [13], [16], [17], [18]
	Knowledge	
	Experience	
	Negotiation	
	Collaboration and communication	
	Stakeholder satisfaction	
	Technical support(tools compatibility/ technology stack)	
Company culture	Industry sector/ Type	[1], [3], [4], [6], [7], [13], [18]
	Company Size	
	Industry-specific standards and regulations	
	Company structure	

B. *Key factors of SQA practices towards Agile IS projects*

The following table presents the identified key factors of SQA practices for Agile IS projects.

TABLE 2. KEY FACTORS OF SQA PRACTICES TOWARDS AGILE IS PROJECTS

Categories	Factors	Research source
Project	Project type	[1], [2], [6], [7], [8], [9], [10], [11], [12], [13], [14], [16], [18]
	Project size	
	Project complexity	
	Project timeline	
	Project cost	
	Project scope	
	Resource availability	
Stakeholder	Requirement	[1], [5], [6], [8], [9], [12], [13], [14], [16], [17], [18]
	Knowledge	
	Experience	
	Negotiation	
	Collaboration and communication	
	Stakeholder satisfaction	
	Technical support(tools compatibility/ technology stack)	
Company culture	Industry sector/ Type	[1], [3], [4], [6], [12], [13], [16], [17], [18]
	Company Size	
	Industry-specific standards and regulations	
	Company structure	

III. METHODOLOGY

The study used a positivist research philosophy, attempting to find empirical facts by direct observation. The deductive approach was used, and data for quantitative analysis were gathered via a survey. It is found that the mono-method, quantitative research strategy is the best fit for this study. This study used a mixed-method approach to data collecting, using both primary and secondary approaches. The study defined the target population as BOI registered software companies in Sri Lanka. The study indicated that cluster sampling was the most appropriate method.

Figure 1 illustrates a conceptual framework; the dependent variable is "SQA practices towards Waterfall and Agile IS projects". There are three independent variables identified, which are, project, stakeholder and company culture.

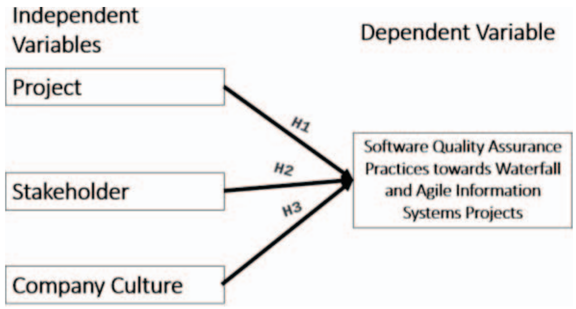

FIGURE 1. CONCEPTUAL FRAMEWORK

To achieve the conceptual framework, have identified three hypotheses are: (1) Hypothesis 1 identifies a relationship between project and SQA practices towards Waterfall and Agile IS projects, (2) Hypothesis 2 identifies a relationship between stakeholder and SQA practices towards Waterfall and Agile IS projects, (3) Hypothesis 3 identifies a relationship between company culture and SQA practices towards Waterfall and Agile IS projects.

The study employed a multi-stage approach to data collection, surveying 63 BOI-registered software companies in Sri Lanka with six individuals from each company, resulting in 378 respondents, accounting for a 20% no-response rate. A four-stage questionnaire development process was followed, including a literature review, supervisory review, pilot survey with industry experts, and a final survey. The questionnaire includes 7 demographic questions and 36 Likert scale questions, with responses recorded on an ordinal scale, enabling comprehensive analysis of participants' perspectives and insights.

The findings revealed that, based on gender, 40.74% of the respondents are male, while 59.26% are female. Regarding the professional designation of the respondents, 75.40% of respondents are engineers/senior engineers, 19.58% are leads/ architects, and 5.03% are managers and above. In terms of project methodology application by the respondents, 70.38% of QA professionals have adopted Agile methodology, emphasizing its dominance in the field. Only 0.26% continue using the Waterfall methodology, while 29.37% employ a hybrid model, combining Agile and Waterfall. These findings highlight Agile's popularity and the increasing trend of integrating methodologies for enhanced adaptability.

IV. ANALYSIS & FINDINGS

A. Descriptive analysis

Independent and dependent variables analysis

TABLE 3. VARIABLES ANALYSIS OF WATERFALL METHODOLOGY

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
SQA practices towards Waterfall IS Projects	378	100.0%	0	0.0%	378	100.0%
Project	378	100.0%	0	0.0%	378	100.0%
Stakeholder	378	100.0%	0	0.0%	378	100.0%
Company culture	378	100.0%	0	0.0%	378	100.0%

According to table 3, each variable comprises 378 cases, representing 100% of the total. There are no missing values, indicated as 0 cases or 0%.

TABLE 4. VARIABLES ANALYSIS OF AGILE METHODOLOGY

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
SQA practices towards Agile IS Projects	378	100.0%	0	0.0%	378	100.0%
Project	378	100.0%	0	0.0%	378	100.0%
Stakeholder	378	100.0%	0	0.0%	378	100.0%
Company culture	378	100.0%	0	0.0%	378	100.0%

According to Table 4, each variable comprises 378 cases, representing 100% of the total. There are no missing values, indicated as 0 cases or 0%.

B. Inferential analysis

Correlation analysis and hypothesis testing

TABLE 5. CORRELATION ANALYSIS AND HYPOTHESIS TESTING OF WATERFALL METHODOLOGY

		SQA practices towards Waterfall IS Projects	Project	Stakeholder	Company culture
SQA practices towards Waterfall IS Projects	Pearson Correlation	1	.936**	.927**	.908**
	Sig. (2-tailed)		.000	.000	.000
	N	378	378	378	378
Project	Pearson Correlation	.936**	1	.976**	.970**
	Sig. (2-tailed)	.000		.000	.000
	N	378	378	378	378
Stakeholder	Pearson Correlation	.927**	.976**	1	.951**
	Sig. (2-tailed)	.000	.000		.000
	N	378	378	378	378
Company culture	Pearson Correlation	.908**	.970**	.951**	1
	Sig. (2-tailed)	.000	.000	.000	
	N	378	378	378	378

**. Correlation is significant at the 0.01 level (2-tailed).

According to the table 5, Pearson correlation coefficients indicate strong positive correlations, with coefficients ranging from 0.908 to 0.936 (all p < 0.01). There is a 0.936 correlation between "SQA practices towards Waterfall IS projects" and "Project", 0.927 with "Stakeholder" and 0.908 with "Company culture". The independent variables themselves also show strong positive correlations, ranging from 0.951 to 0.976. These findings suggest that as the levels of "Project", "Stakeholder" and "Company culture" increase, there is a corresponding increase in "SQA practices towards Waterfall IS projects."

TABLE 6. CORRELATION ANALYSIS AND HYPOTHESIS TESTING OF AGILE METHODOLOGY

		SQA practices towards Agile IS Projects	Project	Stakeholder	Company culture
SQA practices towards Agile IS Projects	Pearson Correlation	1	.904**	.859**	.542**
	Sig. (2-tailed)		.000	.000	.000
	N	378	378	378	378
Project	Pearson Correlation	.904**	1	.856**	.629**
	Sig. (2-tailed)	.000		.000	.000
	N	378	378	378	378
Stakeholder	Pearson Correlation	.859**	.856**	1	.422**
	Sig. (2-tailed)	.000	.000		.000
	N	378	378	378	378
Company culture	Pearson Correlation	.542**	.629**	.422**	1
	Sig. (2-tailed)	.000	.000	.000	
	N	378	378	378	378

**. Correlation is significant at the 0.01 level (2-tailed).

According to the table 6, Pearson correlation coefficients indicate strong positive correlations, with coefficients ranging from 0.859 to 0.904 (all p < 0.01). There is a 0.904 correlation between "SQA practices towards Agile IS projects" and "Project", 0.859 with "Stakeholder" and 0.542 with "Company culture". The independent variables themselves also show strong positive correlations, ranging from 0.629 to 0.904. These findings suggest that as the levels of "Project", "Stakeholder" and "Company culture" increase, there is a corresponding increase in "SQA practices towards Agile IS projects."

V. CONCLUSION

The general objective of this paper is, "To identify SQA practices towards Waterfall and Agile IS projects of BOI registered software companies in Sri Lanka". The literature review highlighted project, stakeholder, and company culture as key towards identification of SQA practices towards Waterfall and Agile IS projects in the global context, through the in depth literature review conducted. These factors evaluated using correlation analysis to identify their significance in BOI registered software companies in Sri Lanka. The findings indicated a strong link between the identified factors towards SQA practices for Waterfall and Agile IS projects. Accordingly, the identification of SQA practices towards Waterfall and Agile IS projects of BOI registered software companies in Sri Lanka is proceeding in the following manner.

TABLE 7. SQA PRACTICES TOWARDS WATERFALL AND AGILE IS PROJECTS

SQA practices of Waterfall methodology	SQA practices of Agile methodology
Test planning and documentation	Pair programming
Traceability matrix	Refactoring
Version control	Continuous improvement
Defect tracking and reporting	Risk based testing
Configuration management	Regression testing suites
Formal approval gates	Continuous monitoring
Documentation management	Documentation and knowledge sharing
Risk assessment and mitigation	Sprint planning
Formal sign-offs	Story mapping
Adherence to coding standards	Retrospectives
Manual and automated testing	Code review
Internal quality management	Automated testing
	Continues delivery
	Cross-functional teams
	Continues feedback
	Short iteration

By aligning SQA practices with the unique needs of each project methodology, software companies can achieve improved outcomes and contribute to the overall advancement of the industry. These insights provide several advantages:

- Understanding the key aspects enables improved resource allocation, ensuring that time, budget, and staff are used effectively to promote project success.
- Making SQA practices to the identified characteristics can result in more efficient quality assurance processes, less mistakes, and higher project success rates.
- Managers and project leaders can make better informed decisions on methodology adoption and customization, ensuring that the chosen practices are consistent with the company's culture and stakeholder expectations.
- Companies that optimize SQA practices based on these insights might increase their competitiveness in the software industry by delivering higher-quality goods and services.

Based on the findings of this paper has identified SQA practices face distinct challenges in Waterfall and Agile IS projects of BOI registered software companies in Sri Lanka. The linear and sequential pattern in Waterfall projects frequently makes it difficult to accommodate changes and adapt to changing needs, resulting in delayed defect detection and increased costs. Conversely, Agile projects, with their iterative and flexible approach, may struggle with maintaining consistent SQA practices amidst frequent changes and rapid development cycles. Both methodologies can experience challenges related to resource constraints, insufficient testing environments, and alignment with industry standards. These challenges underscore the need for tailored SQA practices that address the specific demands and dynamics of each methodology to ensure effective defect management, adherence to quality standards, and successful project outcomes.

Future studies could focus on specific subsets of the current population or explore different geographical regions. A longitudinal approach would enhance the generalizability of findings, while identifying SQA practices across various methodologies beyond Waterfall and Agile could offer further valuable insights.

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